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Extension of Analytical Wall Functions to Supersonic and Hypersonic Flows

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Abstract
This research focuses on the development of wall functions suitable for the prediction of high-speed compressible flows, using RANS-based models. Wall-functions avoid the need for expensive fine near-wall meshes. The conventional log-law-based wall functions, however, have limitations even in incompressible cases, which are further compounded when applied to high-speed compressible flows. The analytical wall function proposed by Craft (Int. J. Heat Fluid Flow 23:148-160, 2002), which involves the analytical solution of simplified boundary-layer forms of the transport equations for the mean flow variables over the near-wall control volumes, has been successfully used in a range of incompressible flows. In this paper, a compressible flow version of the analytical wall function is proposed, which includes the following modifications: (a) improved variation of the convection terms in boundary-layer forms of the mean flow transport equations over the near-wall cells, (b) inclusion of thermal dissipation terms in the simplified analytical equation for the energy variation over the near-wall cells, (c) a more realistic representation of the thermal dissipation process in the discretized energy equation over the near-wall cells, (d) variable molecular viscosity (due to temperature variations) over the viscous sub-layer. The resultant model has been applied to shock wave/turbulent boundary layer interactions up to Mach numbers of 9 and comparisons are drawn with experimental data and with predictions from the log-law-based wall functions and from the Low-Re Launder and Sharma $k-\varepsilon$ model. The predictions resulting from the use of the compressible version of the analytical wall function are consistently closer to the data than those of other wall functions, and in some instances even better than those of corresponding low-Reynolds-number models. Improvements are especially noticeable in the prediction of the wall heat flux rates, where, in the highest Mach number cases, the log-law wall function generally predicts too low values in the shock interaction region, while the low-Reynolds-number models predict too high heat transfer rates, as a result of over-predicting turbulence levels in regions of extremely rapid near-wall temperature variations.

Keywords Shock wave/turbulent boundary layer interaction · Wall heat flux · Turbulence modelling · Wall function

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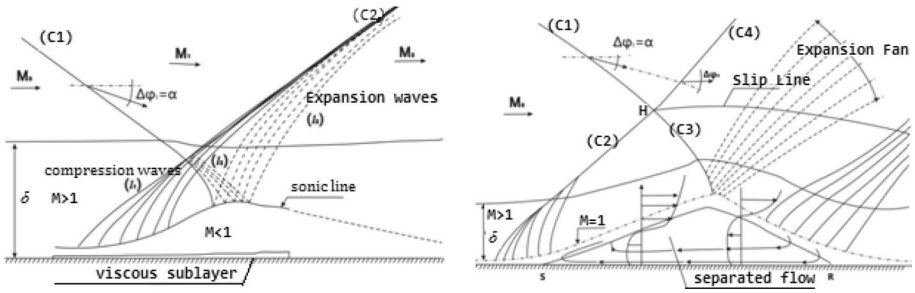


Fig. 1 Shock reflection without (left) and with (right) boundary-layer separation from Délery et al. (1986)

1 Introduction

One of the common occurrences in supersonic flow is shock wave/turbulent boundary layer interaction (SWTBLI), which includes significant changes in temperature, density, velocity and Mach number, over the near-wall regions. One element crucial to the successful use of CFD in the design of advanced aircrafts, especially in complex flow simulations such as those around wing-body junctions, or the inlets of air breathing propulsion systems, where SWTBLIs exist, is the mathematical modelling of turbulence in general and in particular that of near-wall turbulence. A study by Settles and Dodson (1991) examined more than one hundred experimental cases of shock wave interaction with boundary layers, from Mach number 3–13. High-quality data of hypersonic and supersonic SWTBLIs for turbulence modelling are extremely rare, and in their later study (Settles and Dodson 1994) identified only 7 hypersonic and 12 supersonic experiments as being suitable and acceptable for such purposes. Another useful summary of hypersonic flows is provided by Fernholz et al. (1989). A summary of these earlier reviews can be found in Schülein et al. (1996), who themselves were studying an impinging shock case at Mach 5.

The interaction between an impinging-oblique shock and turbulent boundary layer embodies the problems associated with compressibility, flow separation and turbulence, which are all characteristics of SWTBLIs. In this case, the impinging shock is typically generated by a flat-plate or another obstacle with sharp leading edge, and the oblique shock then impinges on (and may be reflected from) a flat wall. For relatively weak interactions (see Fig. 1 left), the boundary layer remains attached, and the interaction is embedded well in the boundary layer. Green (1970) studied cases at Mach 2.5, Re_θ of $4{\times}10^5$, deflection angles of 2°, 2.5° and 5° and showed that the reflected wave is a single shock with equal and opposite reflection angle to that of the incident wave. For stronger interactions (see Fig. 1 right), the oblique shock can be strong enough to cause separation of the turbulent boundary layer. The increase in pressure which takes place at separation generates the leading reflected shock C2. Then shock C2 intersects the oblique shock C1 at point H, from which emanate the two refracted shocks C3 and C4. This pattern, and the point H, can be seen clearly in the Schlieren pictures by Green at deflection angles of 6.5°, 8°, 9.5° and 10.5°.

Another widely examined SWTBLI case is the 2-dimensional compression corner (see Fig. 2). The experiments carried out by Settles et al. (1979), Kuntz et al. (1987), Smits and Muck (1987) are highly cited, mainly because all these papers contain well-defined experimental boundary conditions, adequate documentation data, and a wide range of turning

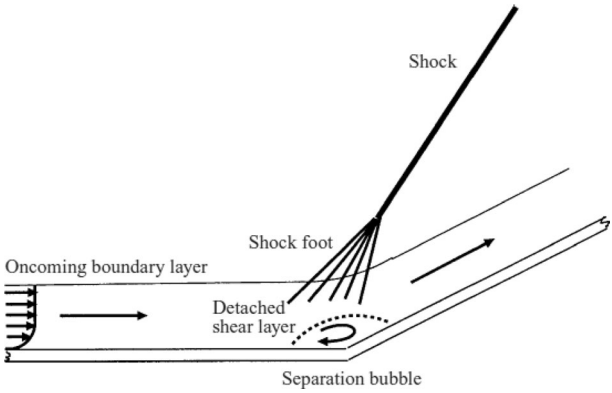


Fig. 2 Sketch of the compression corner flow pattern from Adams (2000)

angles. As an example of this type of application, in this study we consider the experiment at Mach 2.9 and Reynolds number, based on momentum thickness, 75600 by Smits and Muck. Three different corner angles, namely 8°, 16° and 20°, represent flow without separation, flow with incipient separation and flow with significant separation respectively.

Within the Reynolds-averaged modelling approach adopted here, the treatment of the viscous sublayer, which is very close to the solid wall, plays an important role, since molecular diffusion here results in a significant transport of heat and momentum. There are two broad strategies to the near wall turbulence modelling. One widely-used strategy is based on the use of low-Reynolds-number models, summarized by Knight et al. (2003) and Dolling (2001) for shock wave turbulent boundary layer interactions, which requires the use of fine near-wall meshes to resolve the significant changes in velocity, temperature and other flow quantities within the sublayer. Coakley et al. (1994), Huang et al. (1994), Rubesin (1990) and Vuong and Coakley (1987) proposed the introduction of compressibility terms to improve the predictions of flows with SWBLIs. Coakley et al. (1994) and Huang et al. (1994) tested low-Reynolds-number two-equation models such as the $k-\varepsilon$ and $k-\omega$ with compressibility correction terms in either the ε, or the ω transport equations. While in external aerodynamic flow models based on the ω transport equations have become the industry "standard", the comparisons presented in Coakley et al. (1994) and Huang et al. (1994) do not show substantial differences between the predictions of these two models ($k-\varepsilon$ and $k-\omega$). More recently, in a parallel study (Zhang et al. 2022), the authors' group tested low-Re two-equation models such as the $k-\omega$ SST, the linear LS $k-\varepsilon$ and a non-linear version of the $k-\varepsilon$, initially proposed in Craft et al. (1996), and subsequently revised in Craft et al. (2000). In accord with the earlier studies Coakley et al. (1994) Huang et al. (1994), this more recent study also demonstrated that for these supersonic and hypersonic SWBLI flows the quality of the low-Re $k-\varepsilon$ and $k-\omega$ SST predictions is comparable. Use of low-Reynolds-number models requires high resolution of the near-wall regions and thus a significant amount of the computation time can be spent in the resolution of a sublayer covering only around 1% of the domain. In order to overcome this expense, and sometimes convergence difficulties associated with very high aspect ratio near-wall cells, the wall-function approach is widely used in industrial CFD codes, because of its robustness, fast convergence rate, and reasonable accuracy in a number of situations, which are major factors often considered when computing complex geometries.

The standard wall functions are typically based on the log-law and local equilibrium flow conditions, as described by Patankar (1980) and Launder and Spalding (1974). These

forms have been adapted for use in high speed flows, accounting for density variations, and scalings such as that of Trettel and Larsson (2016) have been proposed allowing a range of compressible boundary layer data to collapse onto a log-law shape. However, flows involving strong SWTBLI's are likely to be far from local equilibrium and log-law conditions, and so the use of any log-law based wall function might well be expected to impact on the accuracy of flow predictions. For incompressible flows more refined forms of wall functions have been proposed, ranging from some still based on the log-law (eg Chieng and Launder, 1980), through to others that are not (eg the forms described by Craft et al. 2006). These, however, have not generally been applied to high speed flows. Consequently, in this study, we have developed a compressible flow version of the Analytical Wall Function (AWF) of Craft et al. (2002), which had been been previously successful in the prediction of a range of incompressible flows, and assessed its effectiveness in the computation of high-speed SWBLI flows. Comparisons are drawn with available flow data, and results from standard log-law wall functions and low-Reynolds-number model approaches. We explore how compressibility effects can best be accounted for in such a wall function approach. The objective is to utilize these advanced wall-functions to capture the rapid changes in the turbulence properties in the near-wall region with fast convergence rate, robustness and reasonable accuracy. The open source CFD package OpenFoam v5.0, which employs the finite volume method, is used, since new solvers or utilities can be created within it by users in a fairly convenient manner.

2 Numerical Modelling

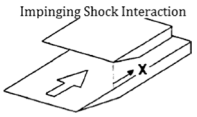
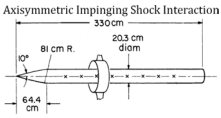
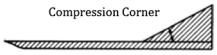
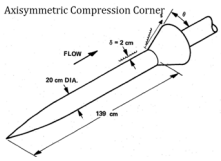
The Favre-Reynolds averaged Navier–Stokes equations are used for the governing equations, which were introduced by Favre (1965).

From the references seven acceptable Shock Wave/Turbulent Boundary Layer Interactions have been selected for the present study as cases with well-defined experimental boundary conditions and adequate documentation of data, which are listed in Table 1, where β denotes the shock generator angle or the ramp corner angle, and θ denotes the boundary layer momentum thickness. The working medium is assumed as a calorically perfect gas as recommended by the research of Roy and Blottner (2006). The equation of state can be taken as $p=\rho RT$. The air properties, such as the gas constant R=287.06$m^2S^{-2}K^{-1}$, specific heat coefficient for constant pressure C_p=1004.06 $J/(kgK)$, molecular Prandtl number Pr=0.7 are used in the calculations, and the Sutherland (1893) equation, $\mu=1.458\times 10^{-6}T^{3/2}/(T+110.3)$, is used to evaluate the dynamic viscosity.

Structured grids are used for all the cases, because of the simple geometries. All grids are uniform in the streamwise direction and refined normal to the wall. This practice ensures that for the high-Reynolds-number turbulence model with wall function approaches, the near-wall cell value of y^+ is about 20 in the regions where the flow is attached. Over a small proportion of the surface, affected by flow separation, this y^+ value rises to as high as 600. For the low-Reynolds-number turbulence models the y^+ value of the near-wall cells, over most of the flow domain, where the flow is attached, is around 0.6.

The original density-based OpenFoam solver, rhoCentralFoam, developed by Greenshields et al. (2010), has been modified with the flux-splitting method by Roe and Pike (1984) using the flux limiter of Van Albada et al. (1997) in the construction of the convection terms. The time integration is carried out using a simple Euler implicit scheme, since it is only marching the flows to a steady state in the present applications.

Table 1 Case details and inflow conditions for supersonic and hypersonic impinging shock interactions and ramp corners

Experimental configuration	References	Ma	$\beta(^\circ)$	θ(mm)	Re_θ
	Reda and Murphy (1973)	2.9	7,10,13	0.82	47000
	Schülein (2004)	5.0	10,14	0.16	5920
	Kussoy and Horstman (1991)	8.2	5,10	0.94	4600
	Kussoy and Horstmann (1975)	7.2	7.5,15	1.25	13600
	Smits and Muck (1987)	2.9	16,20,24	1.3	75600
	Coleman and Stollery (1972)	9.22	15,34,38	–	Re_∞=4.7×10^5 /cm
	Kussoy and Horstman (1989)	7.2	20,30	–	Re_∞=7×10^6 /m

The experimental inflow conditions of impinging shock interactions and ramp corners are listed in Table 1, where Re_θ denotes the Reynolds number based on the momentum thickness θ, and Re_∞ denotes the free-stream unit Reynolds number. The inlet boundary profile in each case is a fully developed turbulent boundary layer, obtained from a preliminary computation of flat plate flow at a specific momentum thickness with 1.5% freestream turbulence intensity and a ratio of turbulent to molecular viscosity μ_t/μ=10 as turbulent inlet conditions. In the upper part of the inlet boundary for the Ma=2.9 impinging shock interaction (as shown in Fig. 3), or on the top boundary for the Ma=5 impinging shock interaction, the oblique shock cuts the boundary, and the inviscid oblique shock relation equations are used to calculate the flow variables downstream of the shock. At the wall, an isothermal no-slip wall boundary condition is used. At top and outlet boundaries, Neumann conditions are used for all components. For the impinging shock interaction cases at Mach numbers of 7.2 and 8.2, the shock generator geometry is also included in the upper part of the domain, in order to produce both the impinging shock from its leading edge and the expansion fan from its rear corner.

Two RANS models are used in the simulations. One of them is the standard $k-\varepsilon$ model, usually referenced to Launder and Spalding (1974), which is a high Reynolds number model. The standard $k-\varepsilon$ model described is only valid for fully turbulent flow, and so is used with the wall function approaches described below. The second model applied is the low-Reynolds-number Launder and Sharma $k-\varepsilon$ scheme (Launder and Sharma 1974) (denoted as LS below), which includes modifications and damping terms for use across the viscous sublayer. The lengthscale correction term proposed by Yap (1987) is also included in the ε equation of the low-Reynolds-number $k-\varepsilon$ model, as it has often been found to be beneficial in avoiding

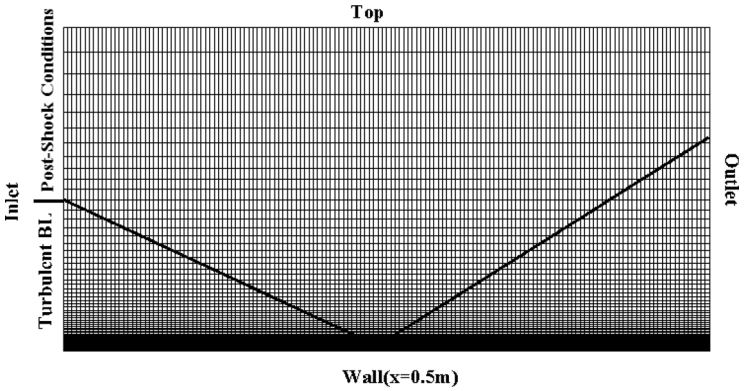


Fig. 3 Computation domain for Ma=2.9 impinging shock interaction

overprediction of turbulence levels in separated flows. In both cases, the turbulent heat fluxes are modelled via a simple gradient diffusion form, with turbulent Prandtl number Pr_t=0.9.

3 Wall Function

Within the Reynolds-averaged modelling approaches, the low-Reynolds-number models use a fine mesh to resolve the viscosity-affected region. In complex geometries, the quality of fine meshes is hard to guarantee in some regions. A more economical alternative, the wall function approach uses a coarse mesh with the high-Reynolds-number models to bridge the gap between the wall and fully-turbulent region. The grids for these two approaches are depicted schematically in Fig. 4. It should be emphasized that the two sample grids of Fig. 4. are used to simply illustrate the two different approaches to near-wall modelling. They are not the grids used in any of the computations presented here.

3.1 Standard Wall Function(SWF)

In local equilibrium turbulent boundary layers, the law of the wall applies. This is used in the standard wall function by assuming a velocity profile of:

$$U^* = \begin{cases} y^* & \left(y^* \le y_L^*\right) \\ \frac{1}{\kappa}\log\left(Ey^*\right) & \left(y^* > y_L^*\right) \end{cases} \tag{1}$$

where U^* and y^* are defined as

$$U^* = U_P C_\mu^{1/4} k_P^{1/2} \Big/ \left(\tau_\mathrm{w}/\rho\right) \quad \text{and} \quad y^* = C_\mu^{1/4} y_P k_P^{1/2} \Big/ v \tag{2}$$

In Eq. (1), κ=0.4178, E=9.793, and C_μ=0.09. The non-dimensional extent of the viscous sublayer, y_L^*, is taken to be 11.04, and k_P and y_P are the values of the turbulent kinetic energy and the wall distance at the near-wall cell centre respectively. The wall shear stress can then be easily obtained as:

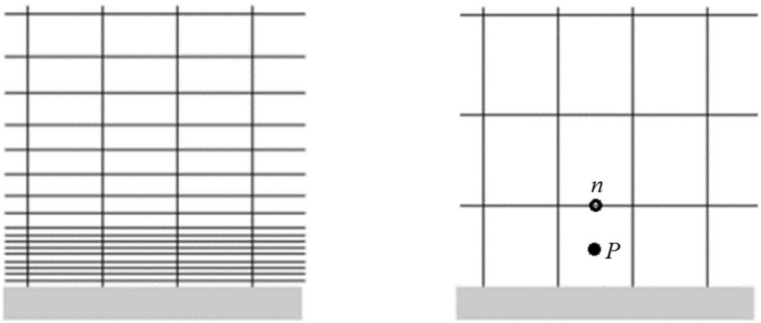


Fig. 4 Near wall grids of near-wall model approach (left) and wall function approach (right)

$$\tau_{\mathrm{w}} = \frac{\rho \kappa U_P k_P^{1/2}}{\log\left(E y_P^*\right)} \qquad \left(y^* > y_L^*\right) \tag{3}$$

For the energy equation, a similar temperature log-law is assumed:

$$T^* = \frac{(T_w - T_P)\rho C_p k_P^{1/2}}{q_w} = \begin{cases} \mathrm{Pr}\, y^* & \left(y^* < y_L^*\right) \\ \mathrm{Pr}_t\left[\frac{1}{\kappa}\log\left(E y^*\right) + P\right] & \left(y^* > y_L^*\right) \end{cases} \tag{4}$$

where q_w is the wall heat flux, C_p is the heat capacity at constant pressure. The P-function by Jayatilleke (1966) is widely used as:

$$P = 9.24\left[\left(\frac{\mathrm{Pr}}{\mathrm{Pr}_t}\right)^{3/4} - 1\right]\left[1 + 0.28e^{-0.007\,\mathrm{Pr}/\mathrm{Pr}_t}\right] \tag{5}$$

The nodal value of the turbulent kinetic energy over the near-wall cells is obtained through the solution of the discretized transport equations for k. Since the production and dissipation of k can both vary significantly across the near-wall cell, the nodal values of these two parameters no longer represent the cell-averaged levels. Cell-averaged generation and dissipation rates for k are thus obtained through analytical integration of simple approximations for how the generation and dissipation vary across the near wall layer. Using assumptions of a simple boundary layer in local equilibrium, including a constant shear stress and a linear turbulence length scale variation across the near-wall cell, the cell-averaged generation and dissipation rates of k over the near-wall cell can be approximated as:

$$\bar{P}_k = \frac{\tau_{\mathrm{w}}^2}{\kappa c_\mu^{1/4} \rho k_P^{1/2} y_n} \log\left(y_n / y_L\right) \tag{6}$$

$$\bar{\varepsilon} = \frac{1}{y_n}\left[\frac{2k_P}{y_L/\nu} + \frac{k_P^{3/2}}{c_l}\log\left(y_{\mathrm{n}}/y_L\right)\right] \tag{7}$$

where the equilibrium length scale constant, c_l, is equal to 2.55, and y_n is the extent of near-wall cell.

3.2 Analytical Wall Function(AWF)

Craft et al. (2002) introduced the AWF based on the analytical solution of the simplified near-wall, boundary layer form of the Reynolds-averaged Navier–Stokes equations, which takes the effects of convection and pressure gradient into account. Writing these simplified momentum and energy transport equations in terms of the wall-parallel and wall-normal co-ordinates x and y respectively, using the effective viscosity and effective diffusivity approximations for the turbulent stresses and heat fluxes respectively, we obtain:

$$\frac{\partial}{\partial y^*}\left[\left(\mu+\mu_t\right)\frac{\partial U}{\partial y^*}\right]=\frac{\mu_w^2}{\rho_w^2 k_P}\left[\frac{\partial(\rho UU)}{\partial x}+\frac{\partial(\rho VU)}{\partial y}+\frac{dP}{dx}\right]_P \tag{8}$$

$$\frac{\partial}{\partial y^*}\left[\left(\frac{\mu}{\mathrm{Pr}}+\frac{\mu_t}{\mathrm{Pr}_t}\right)\frac{\partial T}{\partial y^*}\right]=\frac{\mu_w^2}{\rho_w^2 k_P}\left[\frac{\partial(\rho UT)}{\partial x}+\frac{\partial(\rho VT)}{\partial y}\right]_P \tag{9}$$

where now $y^*=\rho_w y k_P^{1/2}\Big/\mu_w$, and ρ_w, μ_w denote the density and viscosity at the wall. A simple prescription of turbulent viscosity, necessary for the analytical integration of the above equations, is taken as:

$$\mu_t=\begin{cases}0 & \left(y<y_v\right)\\ \mu\alpha\left(y^*-y_v^*\right)=\mu c_\mu c_l\left(y^*-y_v^*\right) & \left(y>y_v\right)\end{cases} \tag{10}$$

where the eddy-diffusivity coefficient, c_μ, is 0.09, so $\alpha=c_\mu c_l$ is 0.2295. The edge of the zero-turbulent-viscosity layer, y_v^*, was determined in Craft et al. (2002), through numerical experiments for fully developed pipe flow, to be 10.8. The subscript P applied to the convective transport and pressure gradient terms on the right hand side of Eqs. (8) and (9) denotes that these terms are assumed to remain constant over the near-wall cell and are evaluated from the nodal values. These simplifications make it possible to solve the above forms of the momentum and energy equation analytically, by integrating them twice with the no-slip and either fixed temperature or wall heat flux boundary conditions at the wall, and boundary conditions of velocity and temperature interpolated from the nodal values at the other face of the near wall cell.

Finally, the wall shear stress can be obtained from

$$\tau_w=-\mu\left.\frac{\partial U}{\partial y}\right|_{y=0} \tag{11}$$

and the wall heat flux is related to the analytical temperature profiles by the following expression:

$$q_w=-C_p\left.\frac{\mu}{\mathrm{Pr}}\frac{\partial T}{\partial y}\right|_{y=0} \tag{12}$$

An approximation for the cell-averaged production of turbulent kinetic energy can be obtained by integrating an expression for the local production of k, based on the analytical velocity variation and the prescribed variation of turbulent viscosity:

$$\overline{P_k} = -\frac{1}{y_n}\frac{\rho\sqrt{k_P}}{\mu}\int_{y_v^*}^{y_n^*} \mu\alpha\left(y^* - y_v^*\right)\left(\frac{\partial U}{\partial y^*}\right)^2 dy^* \tag{13}$$

This term is used for the kinetic energy transport equation to represent the production of the kinetic energy in a consistent way.

The expression of mean dissipation rate across the cell has been also modified. Chieng and Launder (1980), in obtaining Eq. (7) above, had assumed a two-part dissipation profile:

$$\varepsilon = \begin{cases} \frac{2\nu k_P}{y_L^2} & \left(y^* < y_L^*\right) \\ \frac{k^{3/2}}{c_l y} & \left(y^* > y_L^*\right) \end{cases} \tag{14}$$

In order to avoid the discontinuity of the assumed ε profile, Craft et al. (2002) chose a point nearer the wall, y_d, to match the two profile shapes:

$$\frac{k_P^{3/2}}{c_l y_d} = \frac{2\nu k_P}{y_d^2} \quad \text{or} \quad y_d^* = 2c_l = 5.1 \tag{15}$$

The mean dissipation rate can then be obtained by integrating this two-part variation to obtain:

$$\bar{\varepsilon} = \frac{1}{y_n}\left[\frac{2k_P^{3/2}}{y_d^*} + \frac{k_P^{3/2}}{2.55}\ln\left(\frac{y_n}{y_d}\right)\right] \tag{16}$$

As can be seen in a number of earlier publications from the authors' group, this AWF approach has been tested in several incompressible forced, mixed and natural convection flows (Craft et al. 2002, 2014, 2004, 2008b; Katsamis et al. 2022). These tests demonstrate consistent improvements over the log-law-based wall function approach and also greater independence of the resulting predictions from the size of the near-wall cell.

3.3 Modifications to Convection Terms in AWF

First the original version of the AWF, presented above, was introduced into OpenFoam and tested in supersonic flows. The numerical results did show the potential of the AWF in generally predicting reasonably good wall pressure and skin-friction at a considerably lower cost than a low-Re model requires. However, in cases where the SWBLI was strong enough to produce flow separation, such as the case of an impinging shock at Mach 5 and shock deflection angle of 14° shown in Fig. 5, the original AWF formulation tended to produce large unphysical spikes in the skin friction and wall heat flux immediately around the separation point. These spikes were found to be caused mainly as a result of the strong streamwise velocity gradients at the start of separation feeding into the convection terms. However, switching them off entirely would lead to significantly worse results in a number of cases in which the AWF was previously tested, reported by Craft et al. (2008a), and Craft et al. (2014).

In the original implementation of the AWF, the convection terms are evaluated based on nodal values, and treated as being constant across the near wall cells. In an attempt to improve the near-wall modelling of the convection terms we can note that immediately adjacent to the wall one should expect them to increase quadratically with wall distance

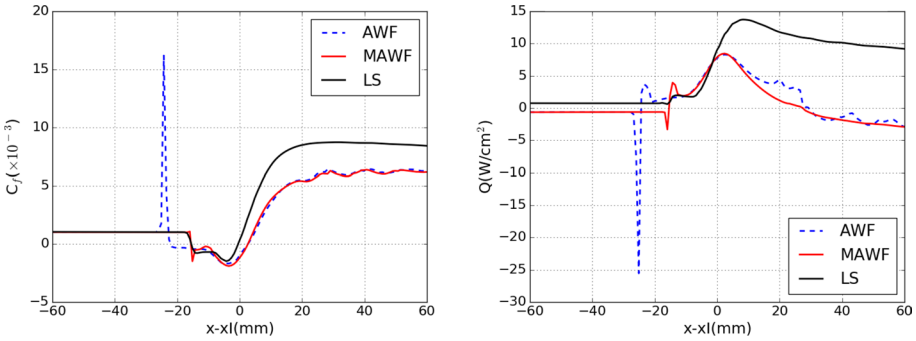


Fig. 5 Wall skin friction and heat flux for Ma=5, β=14° impinging shock interaction predicted by the original form of the analytical wall function, AWF and the modified version, MAWF, in which the convection terms are scaled quadratically with wall distance

(since $U{\sim}y$ and $V{\sim}y^2$ immediately adjacent to the wall). To account for this, the convection terms based on the nodal values have been scaled with the square of the wall distance, allowing Eq. (8) to be rewritten as:

$$\frac{\partial}{\partial y^*}\left[\left(\mu+\mu_t\right)\frac{\partial U}{\partial y^*}\right] = Dy^{*2} + C \tag{17}$$

where D represents the convection terms evaluated from the nodal values and C represents the pressure gradient term from Eq. (8):

$$D = \frac{\mu_w^2}{\rho_w^2 k_P}\frac{1}{y_n^{*2}}\left[\rho U\frac{\partial U}{\partial x} + \rho V\frac{\partial U}{\partial y}\right]_P \qquad C = \frac{\mu_w^2}{\rho_w^2 k_P}\frac{dP}{dx}\bigg|_P \tag{18}$$

The convection terms in the energy equation show a similar near-wall variation (not presented here) as those of the momentum equation. To account for this a similar treatment to that adopted above for the momentum equation has been followed:

$$\frac{\partial}{\partial y^*}\left[\left(\frac{\mu}{\mathrm{Pr}}+\frac{\mu_t}{\mathrm{Pr}_t}\right)\frac{\partial T}{\partial y^*}\right] = \mathrm{D}_{th}y^{*2} \tag{19}$$

where D_{th} represents the convection terms evaluated from the nodal values in the simplified energy Eq.

$$D_{th} = \frac{\mu_w^2}{\rho_w^2 k_P}\frac{1}{y_n^{*2}}\left[\frac{\partial(\rho UT)}{\partial x} + \frac{\partial(\rho VT)}{\partial y}\right]_P \tag{20}$$

The above refinements still allow the simplified momentum and temperature equations to be integrated analytically, and a modified version of the AWF is thus developed in which, over the near-wall control volume, the convective transport of both the wall-parallel momentum and energy is assumed to change quadratically with the wall distance.

To illustrate the impact of the above changes, Fig. 5 shows the predicted skin friction and wall heat flux for the Ma=5, β=14° impinging shock case. The modified AWF (labelled as MAWF) does clearly remove the spikes seen in the predictions of the original AWF formulation.

3.4 Inclusion of Compressibility Effects

The modification to the convection terms terms in the simplified near-wall transport equations does remove the spikes in the AWF predictions of shock-boundary layer interaction flows. As can be seen in the comparisons of Fig. 5, on the other hand, the wall heat flux is still generally lower than that predicted by the Launder-Sharma (LS) model, even in the upstream region. To understand the reasons for this predictive deficiency, it should be noted that the original version of the AWF, developed for incompressible flows, was based on a simplified form of the internal energy equation, whereas most compressible solvers employ the total energy equation:

$$\frac{\partial(\rho E)}{\partial t} + \nabla \bullet \left[\mathbf{u}(\rho E + p)\right] - \nabla \bullet \left(\underline{\sigma} \bullet \mathbf{u}\right) + \nabla \bullet \mathbf{q} = 0 \tag{21}$$

where the stress tensor is expressed as $\underline{\sigma} = \left(\mu + \mu_t\right)\left[\nabla \mathbf{u} + \nabla \mathbf{u}^T - \left(\frac{2}{3}\nabla \bullet \mathbf{u}\right)I\right]$, the total energy is $E = e + \frac{1}{2}|\mathbf{u}|^2$, and the heat flux vector as $\mathbf{q} = -\left(\lambda + \lambda_t\right)\nabla T$.

It was consequently decided to consider extending the use of the total energy equation to the AWF approach. In the original thermal AWF, only the thermal convection terms are involved on the right hand side of Eq. (19). In order to examine the relative size of various terms in the total energy equation, in Fig. 6 the thermal convection terms, the pressure convection and viscous dissipation terms, computed from the LS model in the Ma=5 impinging shock interaction case, are compared over the near-wall region typically covered by the wall function. From the figure, the dissipation term plays a dominant role over the near wall region for high-speed flows.

Order of magnitude analysis, confirmed by interrogation of the numerical solutions, Wang (2018), shows that in the near-wall regions, the elements dominant in the dissipation rate term are the ones which include derivatives of the wall-parallel velocity in the wall-normal direction. Consequently, to account for this in the simplified near-wall energy equation of the AWF, Eq. (19) is adapted to:

$$\frac{\partial}{\partial y^*}\left(\left(\frac{\mu}{\mathrm{Pr}} + \frac{\mu_t}{\mathrm{Pr}_t}\right)\frac{\partial T}{\partial y^*}\right) = \mathrm{D}_{th} y^{*2} - \frac{1}{C_p}\frac{\partial}{\partial y^*}\left(\left(\mu + \mu_t\right)U\frac{\partial U}{\partial y^*}\right) \tag{22}$$

where D_{th} now represents the thermal convection and pressure convection terms in the simplified energy equation:

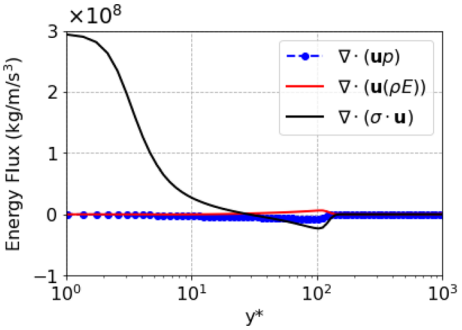


Fig. 6 Comparison of terms in the energy equation over the near-wall region upstream of separation for Ma=5, β=14° impinging shock interaction

$$D_{th} = \frac{\mu_w^2}{C_p \rho_w^2 k_P} \frac{1}{y_n^{*2}} \left[\frac{\partial(\rho UE + Up)}{\partial x} + \frac{\partial(\rho VE + Vp)}{\partial y} \right]_P \tag{23}$$

In the second term on the right hand side of Eq. (22), the wall parallel velocity, U, and its gradient in the wall-normal direction, y, are evaluated using the algebraic functions which result from the analytical integration of the simplified transport equation for the wall-parallel momentum, Eq. (17). Due to the algebraic complexity of this term, the integral of it which arises in the analytical expression for T (see Eqs. 37 and 38 in the appendix) is evaluated numerically.

In addition to the inclusion of the dissipation term in the simplified total energy equation, its representation when solving the equation over the main near-wall grid cell must also be considered and revised. In the standard discretization practice, in common with all other source/sink terms, the viscous dissipation contribution is evaluated at each cell centre. In the main code, the mean total energy dissipation rate term of the near wall cells is thus evaluated as a cell-averaged value, using the analytical velocity distribution, which is consistent with the evaluation of this term in the analytical wall function expression, and is analogous to the way in which cell-averaged forms of the generation and dissipation rates of k are evaluated over the near-wall cells. The cell-averaged viscous dissipation term in the near wall region is constructed as:

$$\overline{\frac{\partial}{\partial y}\left(\left(\mu + \mu_t\right) \frac{\partial U}{\partial y} U \right)} = \frac{\rho_w^2 k_P}{\mu_w^2} \left[\overline{\left(\mu + \mu_t\right) \left(\frac{\partial U}{\partial y^*} \right)^2} + \overline{U \frac{\partial}{\partial y^*} \left(\left(\mu + \mu_t\right) \frac{\partial U}{\partial y^*} \right)} \right] \tag{24}$$

The cell-averaging is done by integrating the terms numerically over the near-wall cell, using the analytical solution for the wall-parallel velocity, U, as a function of the wall distance, y, over each region. In the comparisons that follow, we use CMAWF-D to denote the above modified form of the AWF.

3.5 Accounting for the Variation in Fluid Properties

In the formulations of the AWF tested so far, the transport properties such as density and viscosity have been treated as constant across the near-wall cells. In practice, for hypersonic flows, the rapid changes in near-wall temperature, particularly where strong SWBLIs occur, can lead to significant changes in the fluid properties.

Accounting for variable properties in the AWF involves two decisions. One is deciding where to evaluate the viscosity and density values used in the definition of y^* and also in the model of μ_t, while the other is accounting for the variation of μ when integrating the simplified transport equations across the near-wall layer. In the forms described so far the AWF used wall values of viscosity and density:

$$y^* = \frac{y\sqrt{k_P}}{\mu_w} \rho_w \tag{25}$$

In his study of buoyant flows, where there can also be significant near-wall temperature variations, Gerasimov (2004) found it preferable to prescribe variable fluid properties across the zero turbulent viscosity sublayer, and to make them dependent on the dimensionless

distance, y^*. The temperature at the edge of the zero-turbulent viscosity sublayer, T_v, was calculated from the thermal analytical wall function, and ρ_v and μ_v then obtained from T_v. The dimensionless distance y^* was then defined as:

$$y^* = \frac{y\sqrt{k_P}}{\mu_v}\rho_v \tag{26}$$

and the turbulent viscosity in the turbulent layer modelled as:

$$\mu_t = \mu_v \alpha\left(y^* - y_v^*\right) \tag{27}$$

In order to represent the effects of strong temperature variations on fluid properties across the very near-wall layer, Gerasimov introduced a variable molecular viscosity into Eqs. (8) and (9) across the zero-turbulent viscosity sublayer. In order for the analytical integration of the wall parallel momentum and energy transport equations to remain possible he took the viscosity to be a function of wall distance, and explored the use of linear, hyperbolic and parabolic variation assumptions for the molecular viscosity across the near-wall viscous layer. These same formulations have also been considered in the present work. The linear variation was found to lead to numerical instabilities, but the hyperbolic and parabolic forms led to stable and virtually identical results, as was also the case in Gerasimov (2004). Here, we therefore present only results using the hyperbolic variation:

$$\mu_t = \frac{\mu_v}{1 + b_\mu\left(y^* - y_v^*\right)} \quad \text{where} \quad b_\mu = \frac{\mu_w - \mu_v}{\mu_w y_v^*} \tag{28}$$

The details of the resulting expressions can be found in Appendix A.

In order to test the effect of the above molecular viscosity modifications, Fig. 7 compares distributions of the wall heat flux of Mach 5 and 8.2 impinging shock cases predicted using the compressible form of the CMAWF-D, with and without a variable molecular viscosity in the zero-turbulent viscosity sub-layer and evaluating y^*, and μ_t based on T_v. For the Mach number 5 case, the deviations in the model predictions are fairly small. At a Mach number of 8.2, however, the differences between the predictions using a constant viscosity and those using a variable form are quite significant, because of the stronger near-wall temperature gradients present in this case. The introduction of variable, temperature-dependent, viscosity, and evaluation of y^* and μ_v based on fluid property values at the edge of the viscous sub-layer, results in the prediction of lower levels of skin friction and wall heat flux, bringing the latter much closer to that of the experimental data.

Following the comparisons shown so far, in the more extensive results presented below all the modifications described above are included. These include convective terms scaled with wall distance in the analytical integration of the momentum and energy equations over the near-wall cell, inclusion of compressibility effects in the analytically integrated energy equation and also cell-averaged forms of the compressibility term in the discretized energy equation over the near-wall cells and temperature dependent molecular viscosity over the viscous layer in the analytical integration of the wall-parallel momentum and energy equations, over the near-wall cells. This extended version of the original analytical wall function for compressible flows is now denoted by AWF^{Cmp}.

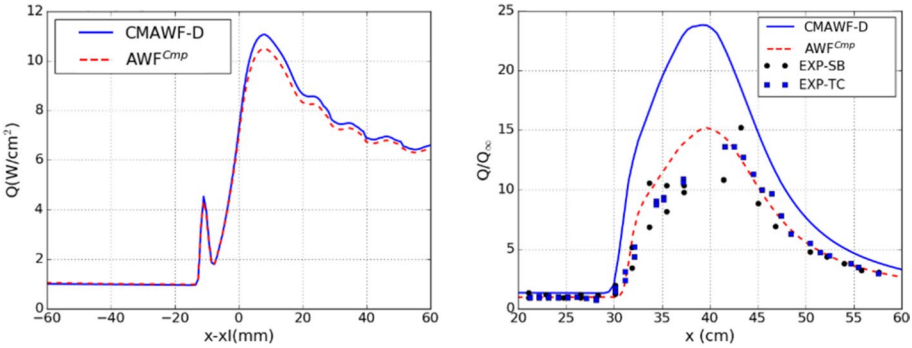


Fig. 7 Wall comparison accounting for fluid variations for Ma=5 (right) and Ma=8.2 (left) impinging shock interactions. CMAWF-D denotes modification of 3.4. AWF^{Cmp} denotes CMAWF-D with variable viscosity across the zero turbulent viscosity sub-layer and evaluating fluid properties at the edge of the viscous sub-layers, as in Eqns (26) and (28)

4 Results

In this section, seven relevant experiments have been selected from the references to be considered in the present study, to assess the effectiveness of the wall functions described above, which are the Ma=2.9, 5, 7.2 and 8.2 impinging shock interactions and Ma=2.9, 7.2 and 9.22 compression corners, summarised in Table 1. As mentioned above, inlet conditions were provided from separate calculations of fully developed boundary layers at the appropriate Mach and Re_θ numbers. Mean velocity profiles from these are compared with the experimental measurements for the Ma=2.9, 7.2 and 8.2 cases in Fig. 8, showing that the proposed wall-function treatment does reproduce the boundary layer profiles over this range of flow conditions.

Grid independence checks have been carried out for all the cases, although for space considerations these are not all reported individually here. As an example, Fig. 9 shows results from different grids for one of the Mach 2.9 impinging shock interaction cases using the low-Re Launder-Sharma (LS) model and high-Re model with wall function approaches. In this case the 200×90 grid was deemed sufficient for the low-Re model and the 150×60 one for the wall functions approaches, which are used for the results presented below. Because here only high-Reynolds-number models have been employed, all the meshes tested have a fixed size for the near-wall control volume. The y^+ value of near-wall

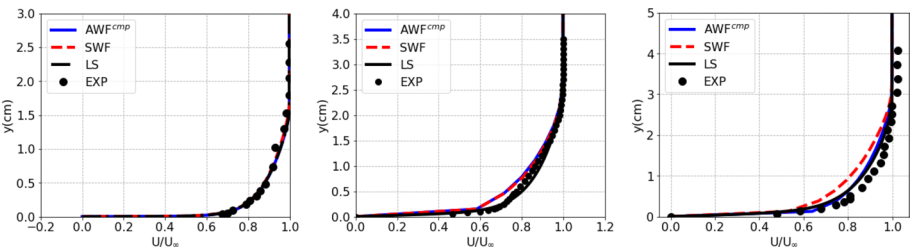


Fig. 8 Near-wall fully-developed velocity comparison for the Ma=2.9 (left) 7.2(middle) and 8.2(right) cases

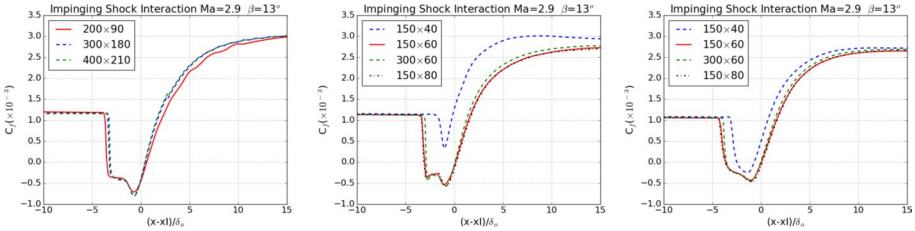


Fig. 9 Grid independence study for the Ma=2.9, β=13° impinging shock interaction predicted by the Launder-Sharma (LS) (left) model, high-Re $k-\varepsilon$ model with AWF^{Cmp} (middle) and SWF (right)

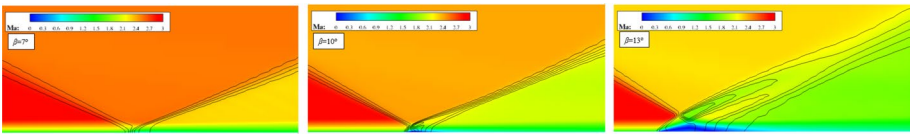


Fig. 10 Iso-lines of mean pressure superimposed on the contours of Mach number for Ma=2.9 impinging shock interaction with shock generator angles of β=7° (left), β=10° (middle), β=13° (right) predicted by the Launder-Sharma (LS) model with the Yap correction

nodes, here defined as $y^+ = \rho_w y C_\mu^{1/4} k_P^{1/2} / \mu_w$, had been found to range from 20 to 600. Details of similar tests for the other cases can be found in Wang (2018).

4.1 Ma=2.9 Impinging Shock Interaction

Figure 10 displays Mach number contours for shock generator angles of 7°, 10° and 13° predicted by the LS model with the Yap correction. As the shock generator angles increase, the impinging shock becomes stronger, the size of the separation zone increases and the separation point advances further upstream.

Figure 11 compares the numerical predictions of wall pressure, skin friction and wall heat flux with experiments for shock generator angles of 10° (left) and 13° (right). For both angles the pressure distribution predicted by both wall function approaches is close to those of the LS and the experimental results. For the skin friction, the log-law-based wall function, SWF, and the compressible flow version of the analytical wall function, AWF^{Cmp}, return very similar distributions. For the most part these are in reasonable agreement with both the LS model predictions and the experimental data. For the 10° case, which is the angle at which flow separation starts to occur, both the experiments and the LS predictions show a small separation region, whereas both wall function approaches predict the flow to be still fully attached. For the wall heat flux, the SWF predicts levels much lower than those resulting from the use of the AWF^{Cmp} and LS in all cases, even upstream of the interaction. The AWF^{Cmp} results match the LS results well in the upstream region. Downstream of the interaction region, the use of the AWF^{Cmp} returns a similar overall distribution of wall heat flux as the LS for both cases, giving substantial improvements in comparison with the log-law-based SWF, which over large sections of the surface returns not only the wrong magnitude, but also the wrong direction, of the wall heat flux.

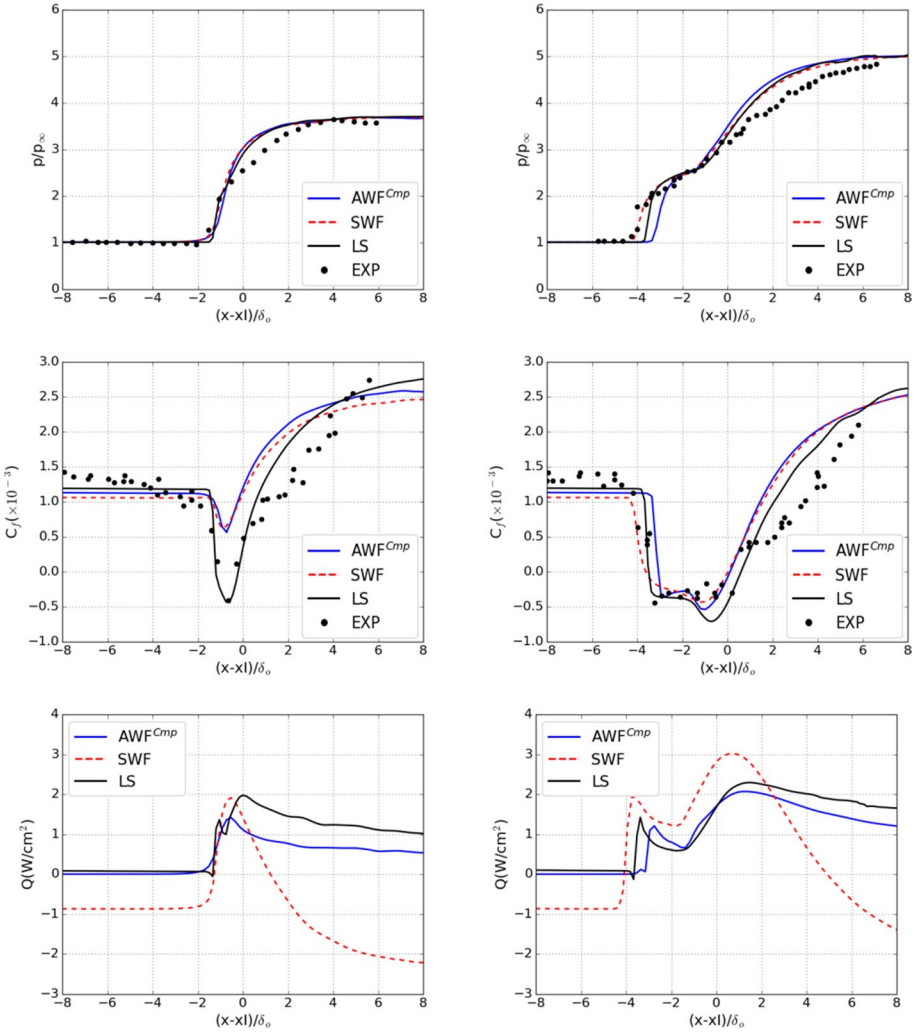


Fig. 11 Surface distributions of wall pressure (top), skin-friction (middle) and wall heat flux (bottom) comparison for Ma=2.9 impinging shock interaction with shock generator angles of 10° (left) and 13° (right) by Reda and Murphy (1973). AWF^{Cmp} denotes high-Re $k-\varepsilon$ model with compressible form of analytical wall function, SWF denotes high-Re $k-\varepsilon$ model with log-law based wall function and LS denotes the Launder-Sharma low-Re $k-\varepsilon$ model with Yap correction

4.2 Ma=2.9 Compression Corner

For this case, following grid independence tests, the results below were obtained using a 280×250 grid for the LS model and a 100×60 mesh for the wall function approaches. Figure 12 displays iso-lines of mean pressure superimposed on Mach number contours at ramp angles of 16°, 20° and 24° by the LS model with the Yap correction. As the ramp angle increases, the separation bubble becomes larger and moves further upstream.

Figure 13 compares the predicted results of wall pressure, skin friction and wall heat flux with those of the experiment. For the wall pressure and friction coefficient, all the approaches to the modelling of near-wall turbulence fit the experimental data reasonably well. The pressure coefficient comparisons suggest that the SWF returns a larger interaction region than the AWF^{Cmp} and LS for the β=20° and 24° cases, but while it agrees well with the experimental measurements for the β=20° case, the AWF^{Cmp} and LS model results fit the experimental data better in the β=24° case. For the skin friction, the AWF^{Cmp} agrees with the LS for all ramp angles. For the 20° cases, the separation region is much smaller than the β=24° case. The SWF shows a tendency to return skin friction levels very slightly lower than those predicted by the LS upstream and downstream of the interaction region. As already noted, both for the pressure coefficient and friction factor, differences among the different predictions are minor. However, in common with the wall heat flux comparisons for the impinging shock cases at the same Mach number, the wall heat flux distributions of Fig. 13 show a greater degree of divergence among the predictions of the three different wall treatments, and especially between the SWF predictions and those of the other two. In this case all three approaches return negative wall heat flux values upstream of the interaction (because of the higher wall temperature employed in this case compared to the Ma=2.9 impinging shock case), although the SWF still returns a rather different overall distribution than the other two approaches.

4.3 Ma=5.0 Impinging Shock Interaction

For the Ma=5 impinging shock interaction, Schülein (2004) has reported surface pressure, skin-friction and wall heat flux measurements, as well as flow visualizations. The computational domain employed here is similar to that of the Ma=2.9 impinging shock interaction case, except that, as noted in Sect. 2, because of the steeper shock angle required in this case, the oblique shock elements are applied along the top boundary (instead of the inlet boundary), in order for the impinging interaction region to occur at the required location. A 240×80 grid is used for the LS model with the Yap correction, while a 120×45 grid is used for the high-Reynolds-number $k-\varepsilon$ model with the wall function approaches.

Figure 14 presents comparisons of the near wall temperature and velocity produced by the LS model and high-Re wall function approaches at the same momentum thickness Re$_\theta$ =0.16 mm in the incoming boundary layer. From the near wall velocity comparison, all modelling approaches return similar velocity distributions at this location, which are in good agreement with the measurements. For the LS temperature predictions, the dimensionless temperature for a very short distance from the wall initially becomes negative, which is outside the bounds of the two boundary values, before reaching a minimum

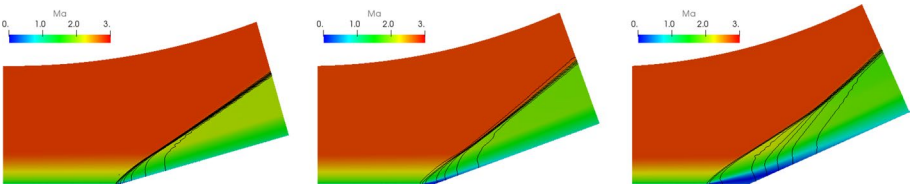


Fig. 12 Iso-lines of mean pressure superimposed on contours of Mach number for Ma=2.9 ramp corner angles of β=16° (left), β=20° (middle), β=24° (right) predicted by the Launder-Sharma (LS) model with the Yap correction

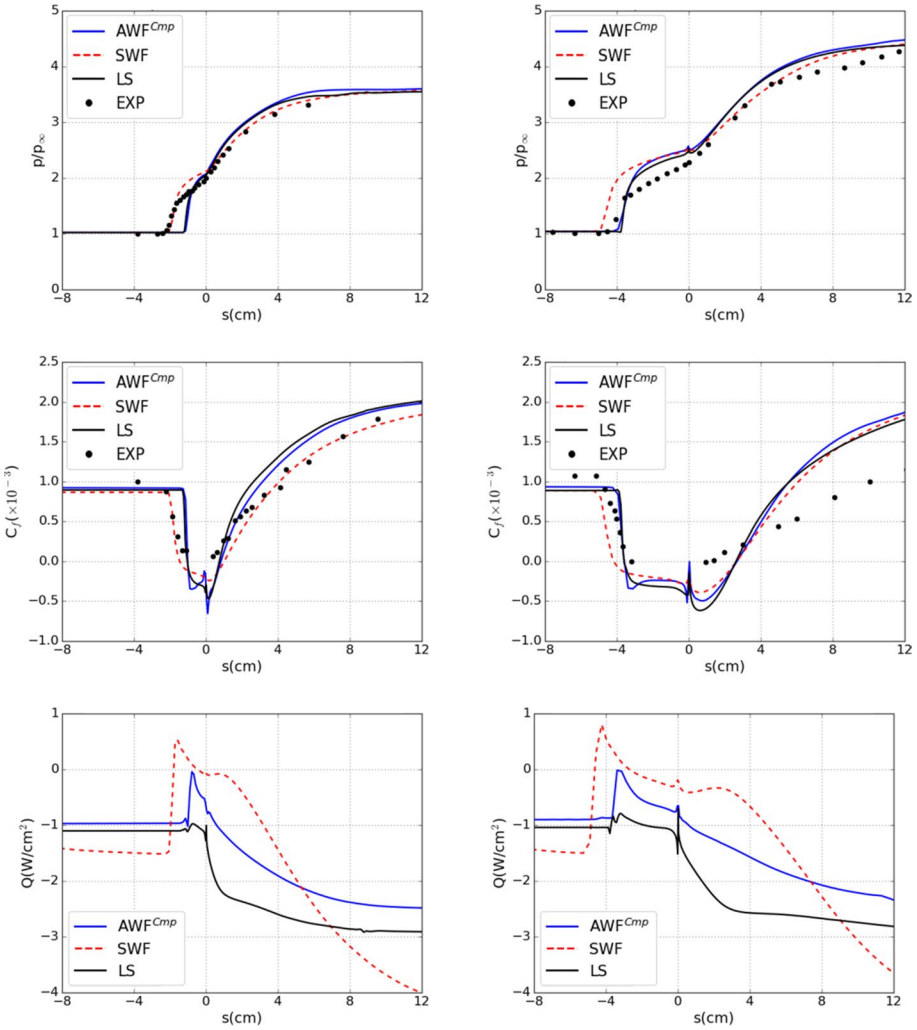


Fig. 13 Surface distributions of wall pressure (top), skin-friction (middle) and wall heat flux (bottom) comparison for Ma=2.9 compression corner with ramp angles of 20° (left) and 24° (right) by Smits and Muck (1987). AWF^{Cmp}, SWF and LS as defined in Fig. 11

point and then changing back to positive. This is due to the compressibility effect, which increases the near-wall gas temperature to levels higher than that of the fixed wall temperature, resulting in a positive wall heat flux. In principle the AWF^{Cmp} approach is able to return such a non-monotonic variation of the temperature across the near-wall control volume, in contrast to the SWF approach. This is revisited in the discussion of Fig. 21.

Figure 15 shows profiles of the mean velocity and temperature at a location downstream of the impingement for the β=14° case. For the velocity, all predictions are in close agreement with the experimental data, which is consistent with the skin friction comparisons of Fig. 16. The corresponding comparisons of the dimensionless temperature show that the measured near-wall profiles are thinner than the predicted ones. The SWF profiles are

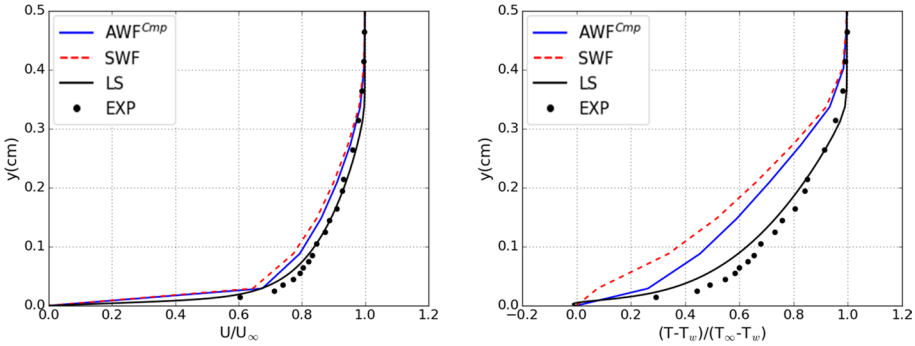


Fig. 14 Near-wall velocity (left) and temperature (right) comparison in the Ma=5 inlet boundary layer. AWF^{Cmp}, SWF and LS as defined in Fig. 11

furthest from the measurements, while the AWF^{Cmp} and LS profiles are close to each other and in agreement with the experimental data.

From the surface pressure, skin-friction and wall heat flux comparisons in Fig. 16, all the numerical results delay the onset of flow separation when compared with the experimental data. The fact that all models return similar predictions of the pressure coefficient suggests that the delayed prediction of flow separation is more likely to be caused by the use of the effective-viscosity approximation rather than the near-wall model. The friction factor comparisons show similar predictive deficiencies as the pressure plots, as far as the onset of flow separation is concerned. Downstream, all models return the correct trend, but both high-Re models under-predict the rise in skin friction by between 20% to 30% depending on the angle. For the two shock generator angle cases shown, measurements of the wall heat flux were also reported, and from the comparison of these it is evident that, once more, for the wall heat flux there are greater deviations among the predictions. The LS and AWF^{Cmp} models predict the correct qualitative variation in Stanton number, but as the ramp angle increases, they start to underestimate the downstream *St* levels, with the AWF^{Cmp} returning progressively lower levels than the LS. The SWF predictions, on the

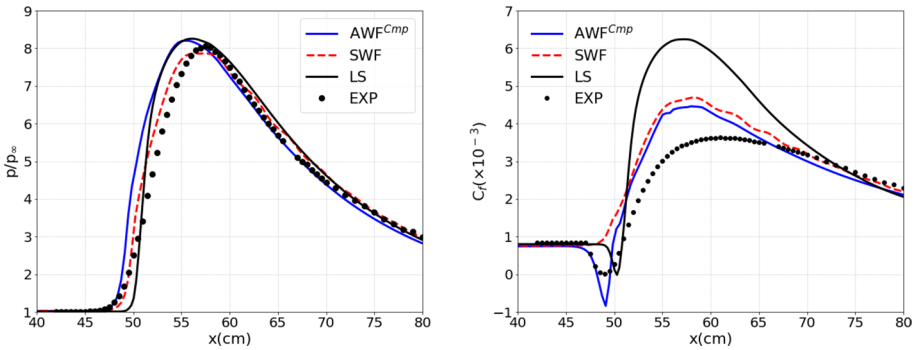


Fig. 15 Near-wall velocity (left) and temperature (right) comparison at a location downstream of the Ma=5 β=14° impinging shock interaction predicted by the LS model and wall function approaches. AWF^{Cmp}, SWF and LS as defined in Fig. 11

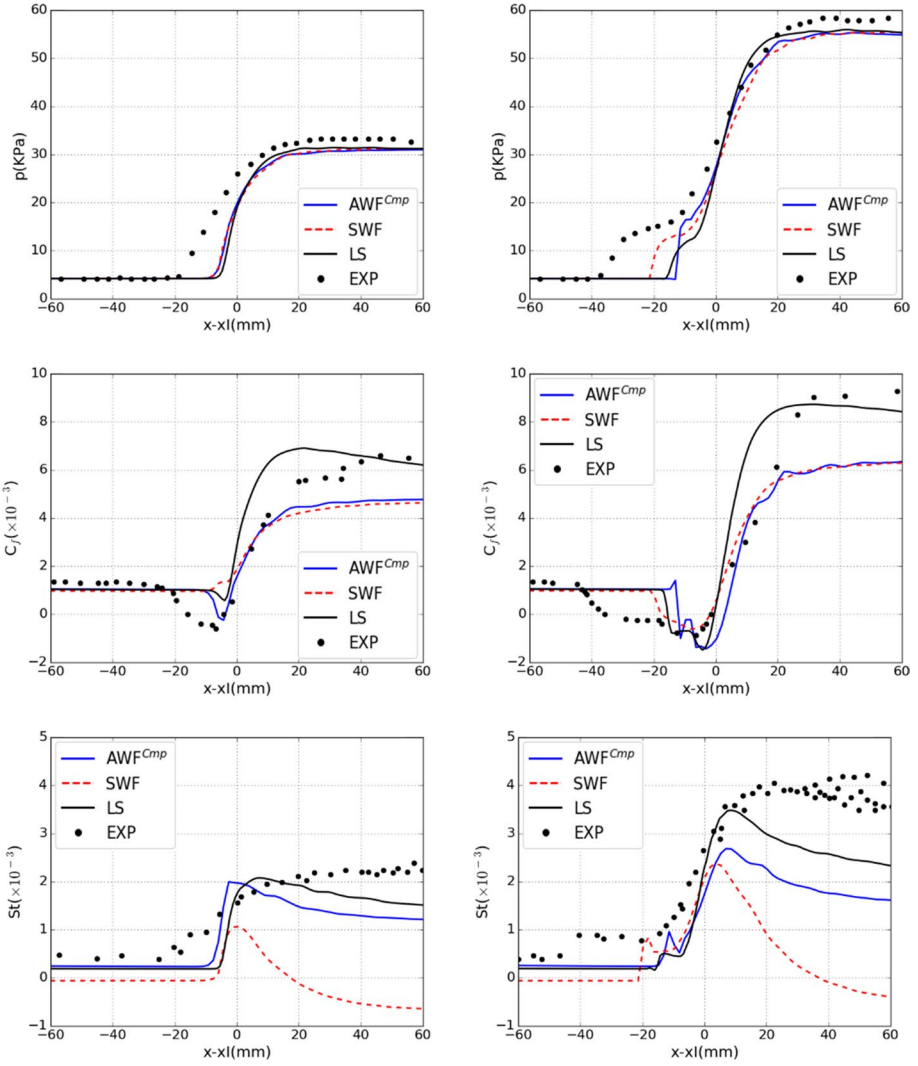


Fig. 16 Surface distributions of wall pressure (top), skin-friction (middle) and wall heat flux (bottom) comparison for Ma=5 impinging shock interaction with shock generator angles of 10° (left) and 14° (right). AWF^{Cmp}, SWF and LS as defined in Fig. 11

other hand, deviate more substantially from the experimental data and the predictions of the other two near-wall models, in some regions returning even the wrong heat flux sign.

4.4 Ma=7.2 Axisymmetric Impinging Shock Interaction

The Ma=7.2 axisymmetric impinging shock interaction experiments were conducted by Kussoy and Horstmann (1975) in the Ames 3.5-Foot Hypersonic Wind Tunnel. The experimental configuration is an axisymmetric cylinder with an axisymmetric shock generator. In this case, the shock generator is included in the computation to produce the impinging

shock and the expansion fan. A 420×105 grid is used for the low-Re turbulence model, and a 280×40 one for the wall function approaches.

The predicted variations in the surface pressure, skin-friction and wall heat flux are compared with those measured in Fig. 17. For the 7.5° case, all the numerical approaches predict a similar peak pressure that matches the experimental value. For the skin-friction, the AWF^{Cmp} predicts return the lowest skin friction levels and is thus in closer agreement with the experimental data than any of the other predictions, including those of the low-Re LS model. For the wall heat flux, the AWF^{Cmp}, including all the modifications described above, leads to the prediction of lower peak wall heat flux than the LS model, resulting in closer agreement with the measured variation. The SWF under-predicts the wall heat flux by a substantial degree, over most of the region.

For the 15° case, the AWF^{Cmp} results have some fluctuations in the interaction region. The SWF tends to predict the lowest wall heat flux values throughout the computation domain as was also observed in the previous cases. While in the 15° case agreement between predictions and experimental data is not as close as in the other cases presented here, this seems to be caused by a more generic predictive failure of the effective-viscosity approximation rather than by the near-wall modelling practices, as the summary by Roy and Blottner (2006) which presented results from a number of other authors who used the $k-\varepsilon$ LS, $k-\varepsilon$ Jones-Launder, $k-\varepsilon$ Rodi, and $k-\omega$ Wilcox models, also showed numerical results that returned higher values of peak pressure, skin-friction, and wall heat flux than those of the experimental results.

4.5 Ma=7.2 Axisymmetric Compression Corner

The Ma=7.2 axisymmetric compression corner experiments were undertaken by Kussoy and Horstman (1989) in a 3.5-Foot Hypersonic Wind Tunnel and include measurements of surface pressure, wall heat-transfer distributions and some flow field properties in the undisturbed and the interaction regions. Here, the cases with flare angles of 20° and 30° are chosen for the numerical tests. Georgiadis et al. (2015) have compared the full-geometry case and no-cone geometry with a straight inflow section to match the boundary layer thickness in the experiments. From the numerical results, these two approaches revealed some differences, but the simplified no-cone geometry as shown in Fig. 18 is deemed to be sufficient. The mesh is compressed around the ramp corner to capture the abrupt changes in that region as shown in Fig. 18. For the subsequent comparisons a 240×150 grid with refinement to the ramp wall as shown in Fig. 18 is used for the Low-Re model computation, while a 100×50 grid is used for the wall function approaches.

Figure 19 compares the numerical results of wall pressure, skin friction and wall heat flux with experiments data at the two angles considered. The wall function approaches using a high-Re turbulence model with coarser near-wall grid return a more gradual increase in wall pressure in the interaction region than that captured by the low-Re model and also that present in the measurements. Downstream of the interaction region, in both cases all the predictions converge to the same pressure level, which is not as high as that of the measured pressures. For the skin friction, all the near-wall modelling approaches return similar skin friction levels in the upstream region of the flow domain. From the predicted skin friction it can be seen that both the LS and the AWF^{Cmp} predict a small separation region around the compression corner in both cases, although the AWF^{Cmp} does give a rather stronger negative C_f in the reversed flow region than the LS

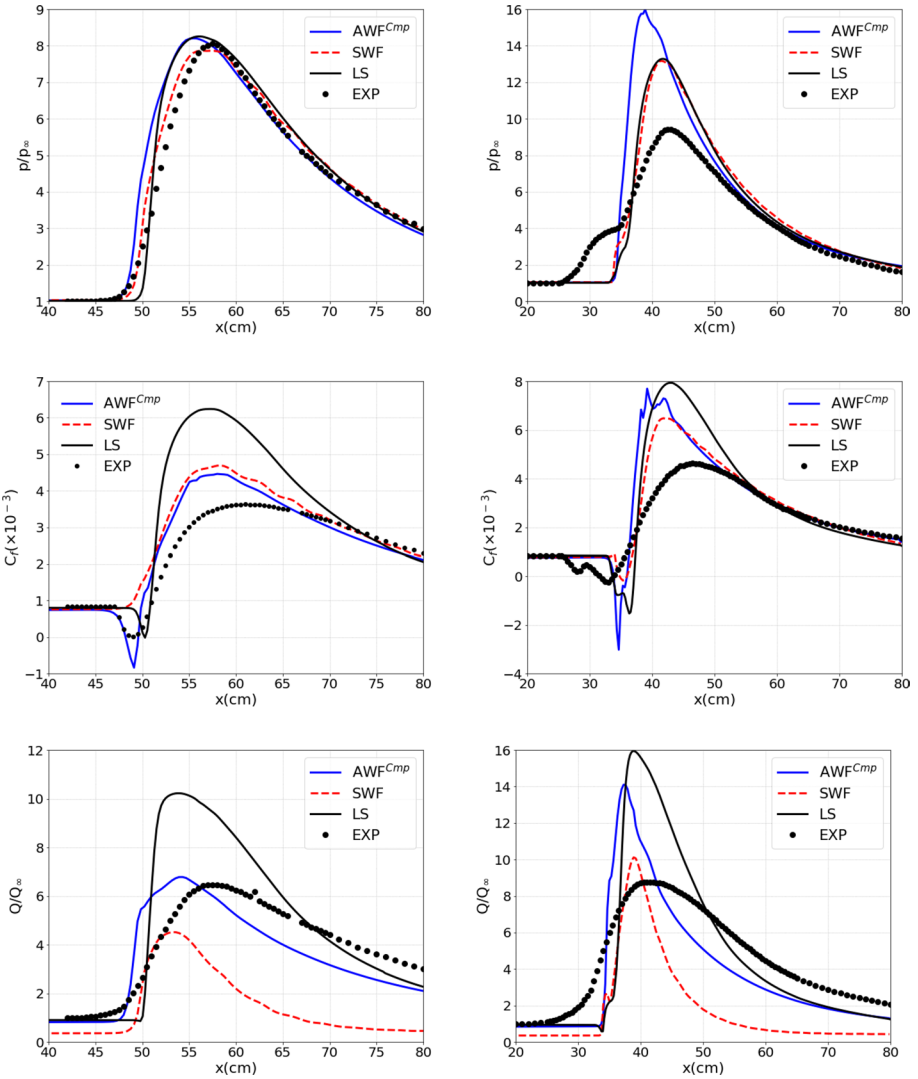


Fig. 17 Near-wall parameters comparison for Ma=7.2 impinging shock interaction with shock generator angles β=7.5° (left) and β=15° (right). AWF^{Cmp}, SWF and LS as defined in Fig. 11

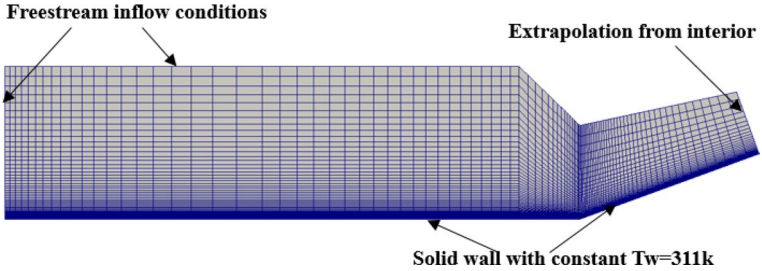


Fig. 18 Computation domain and mesh for Ma=7.2 axisymmetric compression corner

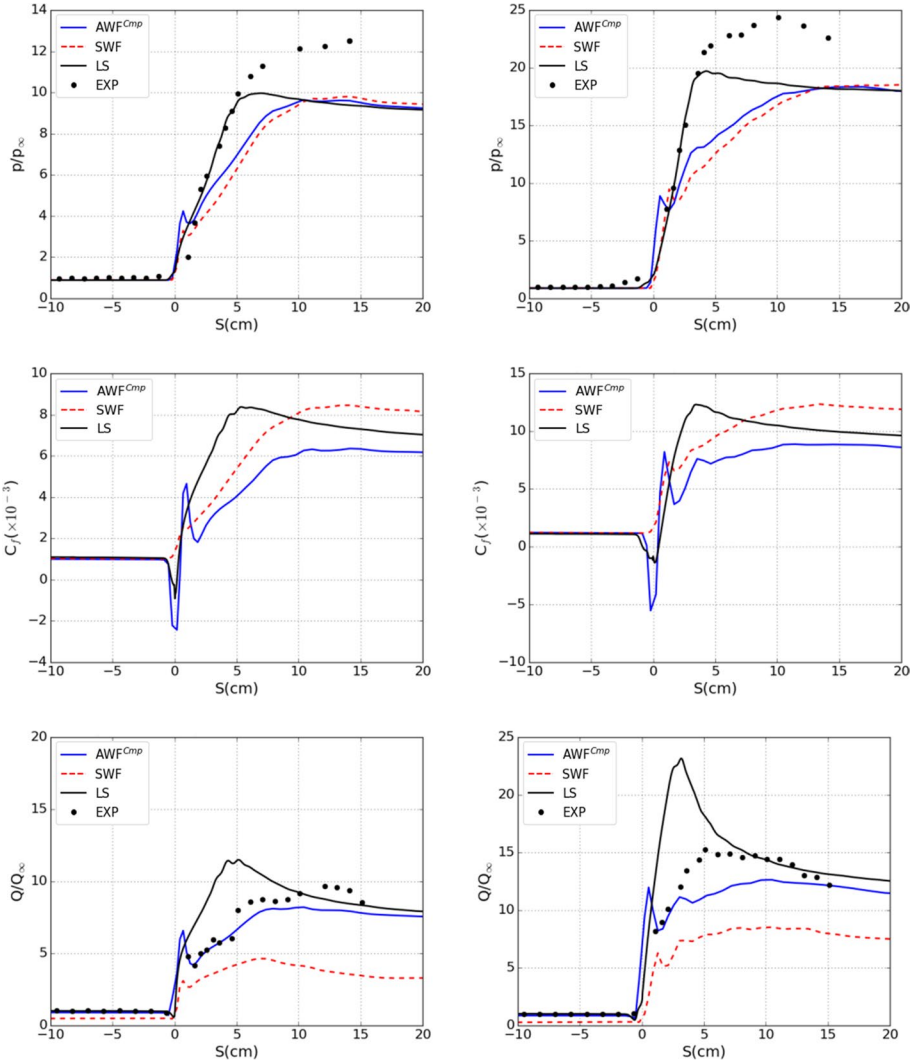


Fig. 19 Near-wall parameters of wall pressure (top), skin-friction (middle) and wall heat flux (bottom) comparison for Ma=7.2 axisymmetric compression corner with ramp angle β=20° (left) and β=30° (right). AWF^{Cmp}, SWF and LS as defined in Fig. 11

model. The SWF does not predict any separation in either ramp angle case. Downstream of the interaction zone, the skin friction predicted by the AWF^{Cmp} is closer to the LS results for both ramp angles.

For the wall heat flux, the low-Re LS model has a tendency to overestimate wall heat flux in the corner region, but returns the correct levels further downstream. The wall heat flux variations predicted by the AWF^{Cmp} follow the experimental pattern more closely. The SWF again fails to predict the wall heat flux accurately throughout most of the flow domain. The wall heat fluxes predicted by the AWF^{Cmp} are similar to those of the experimental data for both cases. Results reported by Georgiadis et al. (2015) using

the $k-\omega$ SST model also return high peak wall heat flux levels which are almost twice as high as those of the experimental data.

4.6 Ma=8.2 Impinging Shock Interaction

The impinging shock interactions at Mach number of 8.2 were investigated experimentally at the NASA Ames Research Centre by Kussoy and Horstman (1991), using a 3.5-ft hypersonic wind tunnel. The mesh employed for the LS model is similar to that used in the Mach=7.2 impinging shock case, with refinements around both the shock generator wall and flat plate wall. For the computations presented here, a mesh of 320×70 is used for the LS model for shock generator angles of 5° and 10° and a 160×60 mesh is used for the wall function approaches.

Figure 20 displays comparisons of surface parameters for which experimental data are available, for the Ma=8.2 impinging shock interaction at shock generator angles of 5° and 10°. Starting with the 5° case, for the surface pressure, all approaches predict wall static pressure variations similar to that of the experiment, with the SWF tending to over-predict the wall pressure in the downstream region. The corresponding wall heat flux comparisons show that at this Mach number, in contrast to the corresponding comparisons at the lower speeds in Fig. 16, the low-Re LS model greatly overestimates wall heat flux after the shock. The log-law-based SWF, and AWF^{Cmp}, on the other hand, are very close to the

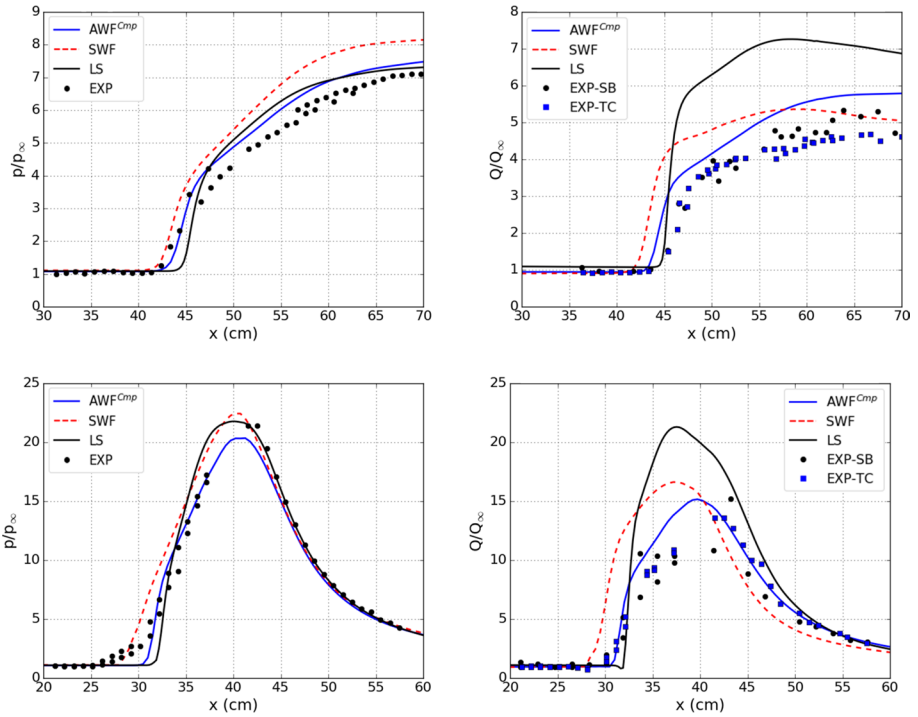


Fig. 20 Near-wall parameters comparison for Ma=8.2 impinging shock interaction with shock generator angle β=5° (top) and β=10° (bottom). AWF^{Cmp}, SWF and LS as defined in Fig. 11

experimental values, with the latter also predicting the location where the wall heat flux increases, more reliably.

For the shock generator angle of 10°, the comparisons lead to conclusions similar to those for the 5° case. The wall pressure variation is again well predicted by all models and if anything agreement between all predictions and the experimental data is closer than for the 5° case. For the wall heat flux, the SWF and the low-Re LS model again over-estimate the wall heat flux over the region of interaction, while the AWF^{Cmp} follows the measured distribution most faithfully.

As noted earlier, in the Ma=5 case, the strong near-wall heating of the fluid, coupled with the fixed wall temperature boundary condition, resulted in a non-monotonic near-wall temperature profile. This is again the case in the even higher Mach number impinging jet cases, as illustrated by Fig. 21, which shows near-wall temperature profiles resulting from the analytical solution of Eqs. (37) and (38) and those predicted by the LS model upstream and downstream of the interaction region for the highest shock angle cases considered, at Mach numbers of 2.9, 5 and 8.2. Upstream of the interaction the LS model predicts a small region of reversed temperature gradient very close to the wall in the Ma=5 case, which is noticeably larger in the Ma=8.2 case; the same is broadly true downstream of the interaction region. The analytical near-wall temperature profiles returned by the AWF^{Cmp} can be seen to mimic the low-Re model behaviour quite closely across all three cases, helping to explain why the AWF^{Cmp} has generally returned a fairly similar behaviour to that of the LS model across the range of cases considered, whilst the SWF, which assumes a near-wall log-law behaviour, has been rather more erratic in its heat-transfer predictions.

4.7 Ma=9.22 Compression Corner

The experiments by Coleman and Stollery (1972) were conducted at a Mach number of 9.22 using the Imperial College no. 2 Gun Tunnel, with nitrogen as the test gas. The nitrogen properties are similar to those of air, and the Sutherland equation is used as:

$$\mu = \frac{1.406 \times 10^{-6} T^{3/2}}{T + 111} \tag{29}$$

For these computations a 240×150 grid is used with the Low-Re model, while a 180×80 grid is used with the wall function approaches.

Figure 22 compares the numerical results of wall pressure and wall heat flux with experiments at ramp angles of β=15°, 34° and 38°. The high-Re $k-\varepsilon$ model with wall function approaches tends to, as also shown at the lower Mach number, Fig. 17, predict a slower increase in pressure in the interaction region at the ramp than either the measurements or LS model. The wall heat flux comparisons at the angle of 15° show that with the low-Re LS model, peak levels at the interaction region are overestimated by around 25%, but lower levels are predicted downstream, while the AWF^{Cmp} predicts the right levels both in the interaction region and downstream. The wall heat flux levels predicted by the SWF are far too low over most of the region.

For the 34° cases, the start of the interaction region moves forward along the flat plate. The LS model returns a generally accurate wall prediction, although does fail to capture the peak shown in the measurements, while the wall function approaches at the initial stages of the interaction region again return a more gradual increase in pressure, but then fall into agreement with the LS prediction. At this angle, use of the low-Re LS results in a far closer prediction of the peak levels of wall heat flux than at the lower angle, but also in a

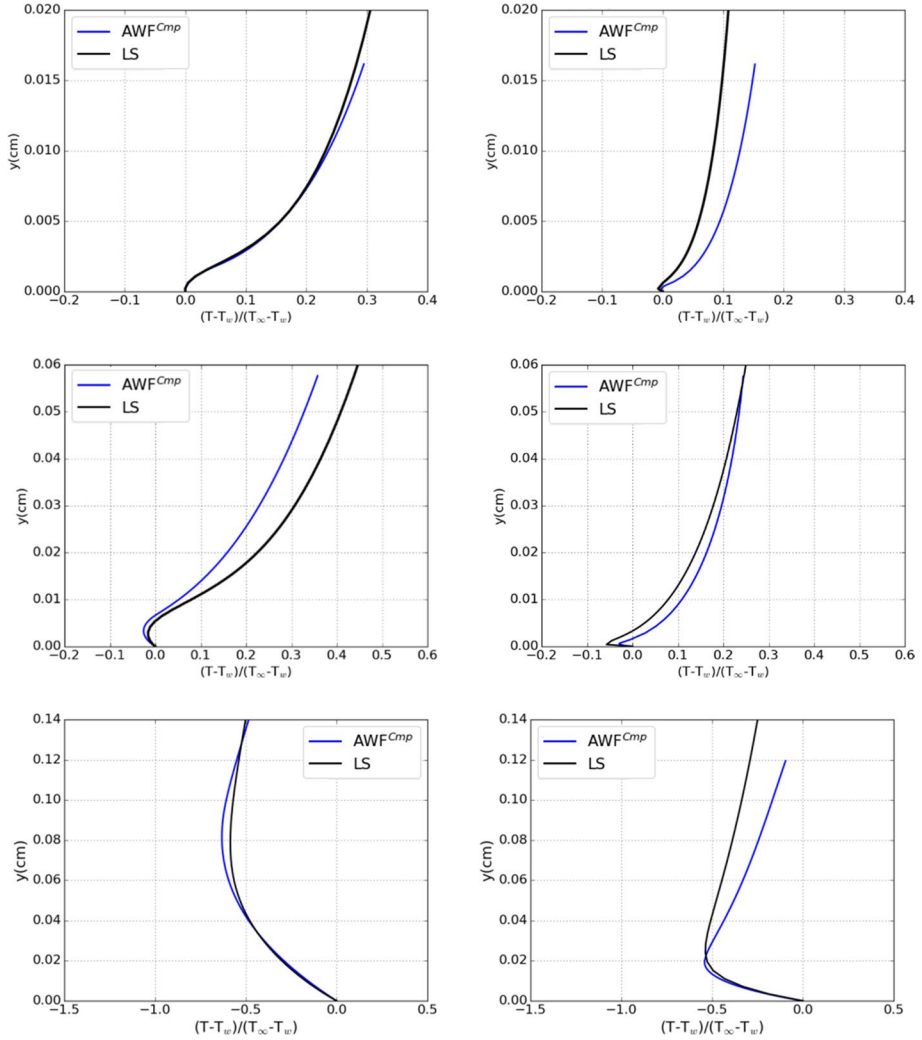


Fig. 21 The analytical temperature, and near-wall Launder-Sharma predicted profiles, upstream (left) and downstream (right) of the interaction zone for the Ma=2.9 (top), 5 (middle) and 8.2 (bottom) impinging shock interactions. AWF^{Cmp} and LS as defined in Fig. 11

slower reduction of the downstream levels than that indicated by the measurements. While underestimating somewhat the peak levels, the AWF^{Cmp} results are on the whole consistently close to the measurements. At this angle the wall heat flux distribution predicted by the SWF is in close agreement with the experimental one, which is in contrast to what was observed at the smaller ramp angle.

For the β=38° case, the measured wall pressure distribution indicates that there is obvious separation around the ramp corner. The LS model under-predicts the extent of separation but reproduces the peak pressure present in the experimental measurements with reasonable accuracy. The wall function approaches fail to predict the sharp pressure peak, but the downstream

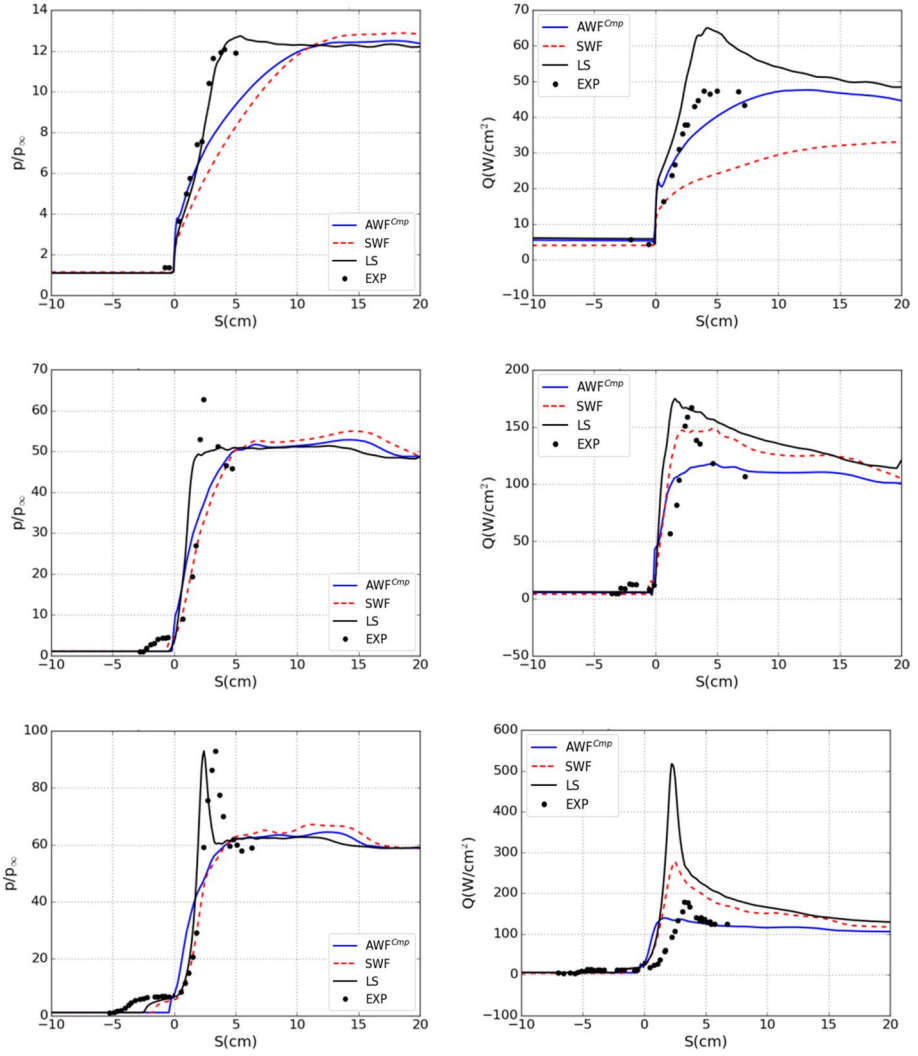


Fig. 22 Wall pressure (left) and wall heat flux (right) comparison for Ma=9.22 compression corner with ramp angle β=15° (top) 34° (middle) and 38° (bottom). AWF^{Cmp}, SWF and LS as defined in Fig. 11

pressure is well reproduced. For the wall heat flux, in contrast to what has been observed at lower angles, the SWF now predicts wall heat flux values 40% higher than in the experiment. Once again the more sophisticated version of the AWF^{Cmp} fits the wall heat flux measurements better than the other two approaches, although, like them, it does predict the increase in wall heat flux to occur rather too steeply and too far upstream compared to the measurements.

5 Concluding Remarks

This study has proposed a number of refinements for the analytical wall function originally proposed by Craft et al. (2002), which extends its applicability to the computation of complex supersonic and hypersonic turbulent flows. The first refinement introduced has been, in the integration of the simplified transport equations, to scale the previously constant over the near-wall cell convective terms with the wall distance. This measure addressed the appearance of spikes observed in numerical predictions using the original analytical wall function around the shock interaction zone.

At higher Mach numbers, close examination of predictions from a low-Re model revealed that compressibility effects increased the temperature of the near-wall gas to levels higher than those of the wall boundary values. These effects necessitated:

(i) The use of the one-dimensional, near-wall version, of the total energy equation with the thermal dissipation terms, for the analytical solution of the temperature over the near-wall cells.
(ii) A more rigorous treatment of the thermal dissipation terms in the solution of the discretized equations for the total energy equation over the near-wall control volumes, similar to the treatment for the generation rate of the turbulent kinetic energy over the same cells.

These changes significantly improved predictions at high Mach numbers.

Finally, another feature of hypersonic flows is the steep variation in the gas near-wall temperature, which causes similarly strong changes to the gas properties such as viscosity. This has been addressed through the inclusion of a temperature dependent molecular viscosity across the layer within which the analytical wall function approach assumes zero turbulent viscosity.

The effectiveness of this extended form of the analytical wall function, referred to as AWF^{Cmp}, has been assessed over Mach number values from 2.9 to 9.22 for impinging shock interactions and compression corners.The extensive comparisons presented here aim to demonstrate the ability of high-Reynolds-number RANS models to cope with some of the features present in turbulent hypersonic flows.

1. Up to Mach number levels of around 2.9 all models perform reasonably well in the prediction of pressure coefficient, but show some deviations from the measurements in the skin friction.
2. At Mach numbers of around 5, all models predict late separation (probably because they all adopt the eddy viscosity approximation), but the LS predicts skin friction distributions which are closer to those measured than the ones of the WF approaches. Nevertheless, all models predict similar C_f distributions and there are only modest differences between the low-Reynolds-number LS and high-Re models. Greater deviations are observed in the prediction of the wall heat flux. The LS and AWF^{Cmp} predictions are in qualitative agreement with the data, with those of the LS being the closest. The log-law-based wall function, SWF, because it is insensitive to compressibility effects, results in wall heat flux predictions which are not even in qualitative agreement with the measured data.
3. At Mach numbers 7 or higher, further deviations are observed between predictions and experimental data. The wall static pressure continues to be predicted with reasonable

accuracy by all models, but with the Low-Re LS model now showing tangible predictive advantages. The skin friction and wall heat flux comparisons again show greater deviations. Most models reproduce the qualitative behaviour, but when it comes to quantitative comparisons the low-Re LS model over-estimates the wall heat flux levels by the greatest amount, while the analytical wall function version with the temperature-dependent molecular viscosity, AWF^{Cmp}, returns levels closest to those of the experiments over the entire range of cases tested at these high Mach numbers. The log-law-based wall function tends to be more erratic, sometimes returning skin friction and wall heat flux levels close to those of the AWF^{Cmp} and sometimes failing to return even the correct qualitative variation.

As suggested by the above summary of the comparisons, this form of the analytical wall function, extended for compressible flows, enables the high-Reynolds-number version of the $k-\varepsilon$ model to return predictions of velocity, temperature and wall parameters which are closer to those of the low-Reynolds-number effective viscosity models than the corresponding predictions resulting from the use of the log-law-based wall function. In some of the cases examined, the wall heat flux distribution returned by the AWF^{Cmp} is closer to that returned by the experimental data than that of the Low-Re model. Since low-Re models have far greater mesh requirements for the resolution of the near-wall regions, the proposed approach can achieve considerable computational saving without the loss in predictive effectiveness which results from the use of the log-law-based wall function. As established in Craft et al. (2008a) and Craft et al. (2008b), one of the advantages of the AWF strategy over the conventional log-law-based wall-function approach, is that it can account for boundary layer three-dimensionality. Furthermore, Gerasimov (2004) found that the original AWF formulation was fairly insensitive to the near-wall cell size over a substantial range of y^+ values, which could equally be expected of the present version, making it more reliable than a standard wall function approach in complex applications where the near-wall grid sizing may be difficult to control. Employing the proposed approach in computations of three-dimensional hyper-sonic flows, with the three-dimensional framework adopted in Craft et al. (2008a), will result in even greater computational savings. Moreover, combining this approach with more advanced turbulence models, not based on the effective-viscosity approximation, should result in further advances in modelling fidelity. In future, as more data become available on the real-gas effects, it will become possible to further extend to account for these phenomena.

Appendix

a. Hydrodynamic Analytical Wall Function

In all subsequent equations, subscript "1" denotes the zero-turbulent viscosity sublayer ($y^* < y_v^*$), and subscript "2" the turbulent region of the turbulent layer ($y^* > y_v^*$).

The second integration of Eq. (18) with hyperbolic variation for the molecular viscosity over the zero turbulent viscosity sub-layer gives:

$$U_1 = \frac{1}{\mu_v}\left(\begin{array}{l}\frac{D_1 y^{*4}}{12} + \frac{C_1}{2}y^{*2} + A_1 y^* \\ +b_\mu\left(\frac{D_1 y^{*5}}{15} - \frac{D_1 y_v^*}{12}y^{*4} + \frac{C_1}{3}y^{*3} + \frac{(A_1 - C_1 y_v^*)}{2}y^{*2} - A_1 y_v^* y^*\right)\end{array}\right) \tag{30}$$

$$U_2 = \frac{1}{\mu_v}\begin{bmatrix} \frac{D_2}{3}\frac{1}{\alpha^3}\left[\frac{\alpha^2}{3}y^{*3} - \frac{\alpha\left(1-\alpha y_v^*\right)}{2}y^{*2} + \left(1-\alpha y_v^*\right)^2 y^* - \left(1-\alpha y_v^*\right)^3\frac{1}{\alpha}\ln Y\right] \\ +C_2\frac{1}{\alpha}\left[y^* - \left(1-\alpha y_v^*\right)\frac{1}{\alpha}\ln Y\right] + A_2\frac{1}{\alpha}\ln Y + B_2 \end{bmatrix} \tag{31}$$

From the near wall cell boundary conditions, the coefficients in the above equations are:

$$A_1 = \frac{\mu_v U_n - N}{\frac{\ln Y_n}{\alpha} + y_v^* - 0.5 b_\mu y_v^{*2}} \tag{32}$$

$$A_2 = \left(D_1 - D_2\right)\frac{y_v^{*3}}{3} + \left(C_1 - C_2\right)y_v^* + A_1 \tag{33}$$

$$\begin{aligned} B_2 = {} & \frac{D_1 y_v^{*4}}{12} + \frac{C_1}{2}y_v^{*2} + A_1 y_v^* - b_\mu\left(\frac{D_1}{60}y_v^{*5} + \frac{C_1}{6}y_v^{*3} + \frac{1}{2}A_1 y_v^{*2}\right) \\ & - \left[\frac{D_2}{3}\frac{1}{\alpha^3}\left[\frac{\alpha^2}{3}y_v^{*3} - \frac{\alpha\left(1-\alpha y_v^*\right)}{2}y_v^{*2} + \left(1-\alpha y_v^*\right)^2 y_v^*\right] + C_2\frac{1}{\alpha}y_v^*\right] \end{aligned} \tag{34}$$

The coefficient N in (34) represents

$$\begin{aligned} N = {} & \frac{D_2}{3}\frac{1}{\alpha^3}\left[\frac{\alpha^2}{3}y_n^{*3} - \frac{\alpha\left(1-\alpha y_v^*\right)}{2}y_n^{*2} + \left(1-\alpha y_v^*\right)^2 y_n^* - \left(1-\alpha y_v^*\right)^3\frac{1}{\alpha}\ln Y_n\right] \\ & + C_2\frac{1}{\alpha}\left[y_n^* - \left(\frac{1}{\alpha} - y_v^*\right)\ln Y_n\right] + \frac{1}{\alpha}\ln Y_n\left(\left(D_1 - D_2\right)\frac{y_v^{*3}}{3} + \left(C_1 - C_2\right)y_v^*\right) \\ & + \left(\frac{D_1 y_v^{*4}}{12} + \frac{C_1}{2}y_v^{*2} - b_\mu\left(\frac{D_1}{60}y_v^{*5} + \frac{C_1}{6}y_v^{*3}\right)\right) \\ & - \left[\frac{D_2}{3}\frac{1}{\alpha^3}\left[\frac{\alpha^2}{3}y_v^{*3} - \frac{\alpha\left(1-\alpha y_v^*\right)}{2}y_v^{*2} + \left(1-\alpha y_v^*\right)^2 y_v^*\right] + C_2\frac{1}{\alpha}y_v^*\right] \end{aligned} \tag{35}$$

where

$$Y = 1 + \alpha\left(y^* - y_v^*\right) \tag{36}$$

b. Thermal Analytical Wall Function

In the zero-turbulent viscosity sublayer ($y^* < y_v^*$), the second integration of Eq. (22) with hyperbolic variation gives:

$$T_1 = \frac{\mathrm{Pr}}{\mu_v}\begin{bmatrix} \frac{\mathrm{D}_{th1}y^{*4}}{12} + A_{th1}y^* + b_\mu\left(\frac{\mathrm{D}_{th1}y^{*5}}{15} - \frac{\mathrm{D}_{th1}y_v^*}{12}y^{*4} + \frac{A_{th1}}{2}y^{*2} - A_{th1}y_v^* y^*\right) \\ -\int_0^{y^*}\frac{1}{C_p}\left(1 + b_\mu\left(y^* - y_v^*\right)\right)\mu U_1\frac{\partial U_1}{\partial y^*}dy^* \end{bmatrix} + T_w \tag{37}$$

In the turbulent layer ($y^* > y_v^*$), the analytical temperature is:

$$T_2 = \frac{\Pr}{\mu_v}\left\{\begin{array}{l}\frac{D_{th2}}{3}\frac{1}{\alpha_t^3}\left[\frac{\alpha_t^2}{3}y^{*3} - \frac{\alpha_t\left(1-\alpha_t y_v^*\right)}{2}y^{*2} + \left(1-\alpha_t y_v^*\right)^2 y^* - \left(1-\alpha_t y_v^*\right)^3\frac{1}{\alpha_t}\ln Y_T\right] \\ +\frac{A_{th2}}{\alpha_t}\ln Y_T + B_{th2} - \frac{1}{\mu_v C_p}\int_{y_v^*}^{y^*}\frac{1}{Y_T}\mu_v U_2\left(\mu_v Y\frac{\partial U_2}{\partial y^*}\right)dy^*\end{array}\right\} \tag{38}$$

where the coefficients in the above equations are

$$B_{th2} = A_{th1}y_v^* - \frac{1}{2}b_\mu A_{th1}y_v^{*2} + BT + \frac{\mu_v}{\Pr}T_w \tag{39}$$

$$A_{th2} = \left(\frac{\mathrm{D}_{th1}}{3} - \frac{D_{th2}}{3}\right)y_v^{*3} + A_{th1} \tag{40}$$

$$\begin{aligned} BT = &\frac{\mathrm{D}_{th1}y_v^{*4}}{12} - \frac{\mathrm{D}_{th1}}{60}b_\mu y_v^{*5} - \int_0^{y_v^*}\frac{1}{C_p}\left(1+b_\mu\left(y^*-y_v^*\right)\right)\mu U_1\frac{\partial U_1}{\partial y^*}dy^* \\ &- \frac{D_{th2}}{3}\frac{1}{\alpha_t^3}\left[\frac{\alpha_t^2}{3}y_v^{*3} - \frac{\alpha_t\left(1-\alpha_t y_v^*\right)}{2}y_v^{*2} + \left(1-\alpha_t y_v^*\right)^2 y_v^*\right]\end{aligned} \tag{41}$$

In order to complete the thermal wall function, the final expression for the wall temperature needs to be substituted into the code for prescribed heat-flux boundary conditions. From Eq. (39), when $y^* = y_n^*$, and $T_2 = T_n$:

$$T_w = T_n - \frac{\Pr}{\mu_v}\left\{\begin{array}{l}\frac{D_{th2}}{3}\frac{1}{\alpha_t^3}\left[\begin{array}{l}\frac{\alpha_t^2}{3}y_n^{*3} - \frac{\alpha_t\left(1-\alpha_t y_v^*\right)}{2}y_n^{*2} \\ +\left(1-\alpha_t y_v^*\right)^2 y_n^* - \left(1-\alpha_t y_v^*\right)^3\frac{1}{\alpha_t}\ln Y_{Tn}\end{array}\right] \\ +\frac{\ln Y_{Tn}}{\alpha_t}\left(\left(\frac{\mathrm{D}_{th1}}{3} - \frac{D_{th2}}{3}\right)y_v^{*3} + A_{th1}\right) + A_{th1}y_v^* - \frac{1}{2}b_\mu A_{th1}y_v^{*2} + BT \\ -\frac{1}{\mu_v C_p}\int_{y_v^*}^{y_n^*}\frac{1}{Y_T}\mu_v U_2\left(\mu_v Y\frac{\partial U_2}{\partial y^*}\right)dy^*\end{array}\right\} \tag{42}$$

For the isothermal wall boundary conditions:

$$q_w = -\frac{\rho_v c_p\sqrt{k_P}}{\mu_v}A_{th1} \tag{43}$$

where A_{th1} can be obtained from (42) as:

$$A_{th1} = \frac{1}{\left(\frac{\ln Y_{Tn}}{\alpha_t} + y_v^* - \frac{b_\mu}{2}y_v^{*2}\right)}\left\{\begin{array}{l}\frac{\mu_v}{\Pr}\left(T_n - T_w\right) \\ -\frac{D_{th2}}{3}\frac{1}{\alpha_t^3}\left[\begin{array}{l}\frac{\alpha_t^2}{3}y_n^{*3} - \frac{\alpha_t\left(1-\alpha_t y_v^*\right)}{2}y_n^{*2} \\ +\left(1-\alpha_t y_v^*\right)^2 y_n^* - \left(1-\alpha_t y_v^*\right)^3\frac{1}{\alpha_t}\ln Y_{Tn}\end{array}\right] \\ -\frac{\ln Y_{Tn}}{\alpha_t}\left(\frac{\mathrm{D}_{th1}}{3} - \frac{D_{th2}}{3}\right)y_v^{*3} - BT \\ +\frac{1}{\mu_v C_p}\int_{y_v^*}^{y_n^*}\frac{1}{Y_T}\mu_v U_2\left(\mu_v Y\frac{\partial U_2}{\partial y^*}\right)dy^*\end{array}\right\} \tag{44}$$

where

$$Y_T = 1 + \alpha_t\left(y^* - y_v^*\right) = 1 + \alpha\frac{\mathrm{Pr}}{\mathrm{Pr}_t}\left(y^* - y_v^*\right) \tag{45}$$

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Declarations

Conflict of interest The authors declare that they have no conflicts of interest.

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